Scheduler Evaluation Results - CSL3030

Scheduler Evaluation Results

Date: 2026-11-13. General sched_bench results are reported separately from the focused donation demonstration.

Data and validation

Historical CPUS=1 benchmark CSVs and the new one-pass CPUS=1 and CPUS=3 runs were included only when each policy/workload had exactly three distinct workflow IDs (1-3), valid nonnegative elapsed ticks and a successful run summary. Workflow records are not counted as repetitions. A repetition is one complete policy/workload invocation.

Source directory	CPUs	Complete policy/workload runs
20261112_224846_031022	1	9
20261113_101619_302636	1	9
20261113_102920_364650	1	3
20261113_104756_383436	1	9
20261113_104805_318328	3	9

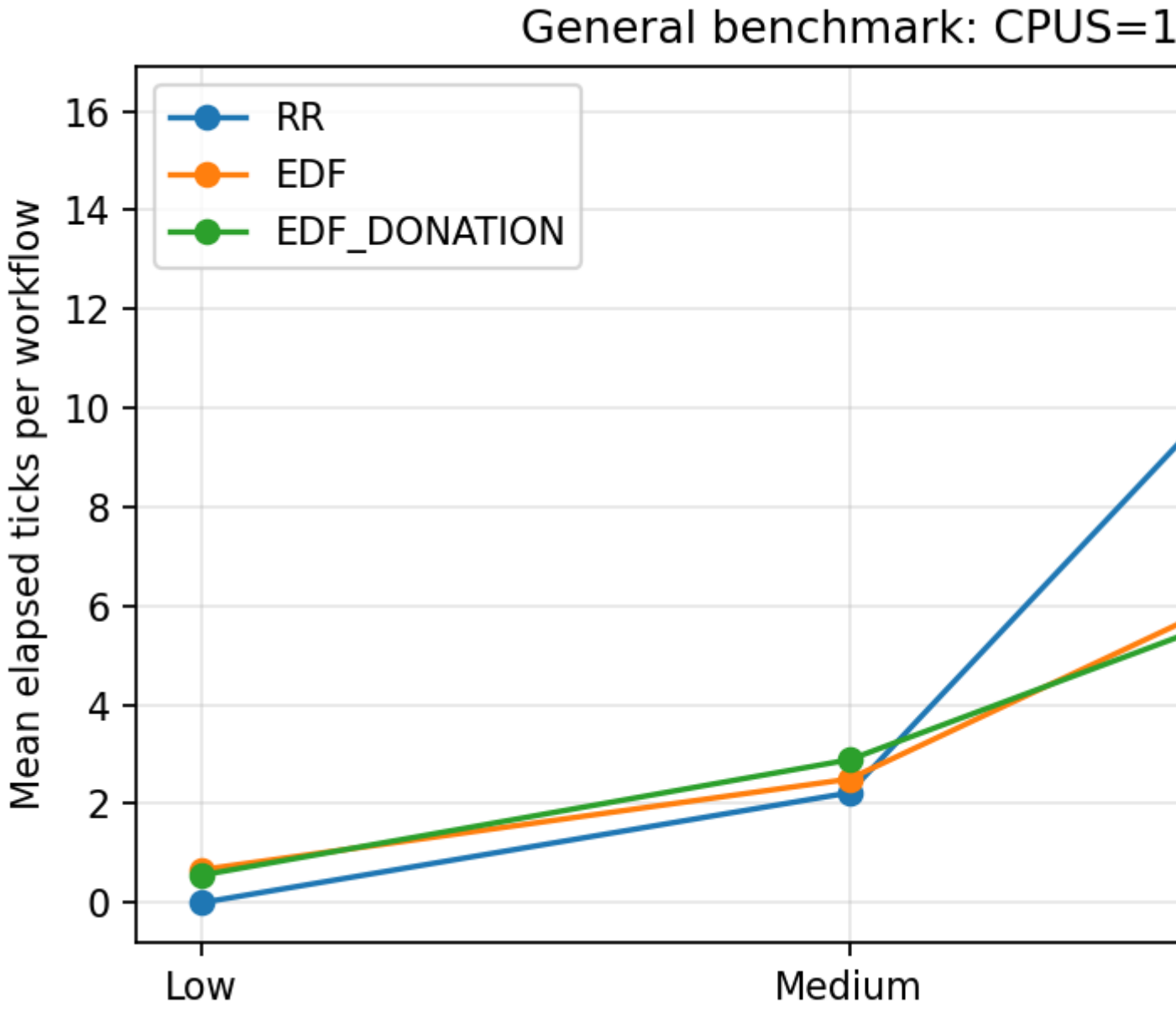
General benchmark: actual repetitions

CPUs	Policy	Low	Medium	High
1	RR	3	3	3
1	EDF	4	4	4
1	EDF_DONATION	3	3	3
3	RR	1	1	1
3	EDF	1	1	1
3	EDF_DONATION	1	1	1

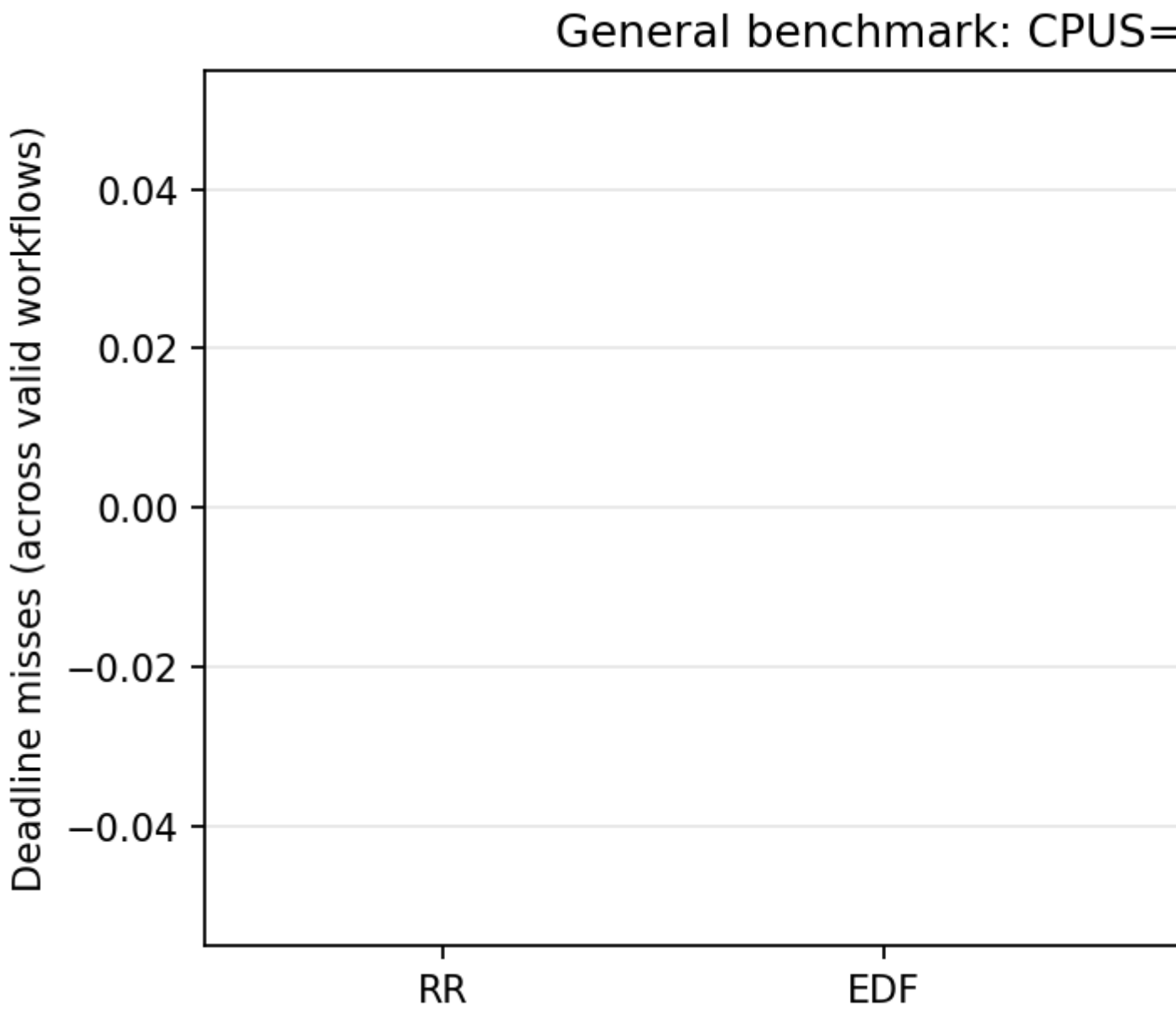
Each valid run contributes three workflow measurements. Miss totals below count misses across those measured workflows; means are per-workflow elapsed ticks.

General benchmark: measured outcomes

CPUs	Policy	Workload	Valid repetitions	Mean elapsed ticks	Deadline misses / workflows
1	RR	Low	3	0.00	0 / 9
1	RR	Medium	3	2.22	0 / 9
1	RR	High	3	16.11	0 / 9
1	EDF	Low	4	0.67	0 / 12
1	EDF	Medium	4	2.50	0 / 12
1	EDF	High	4	8.67	0 / 12
1	EDF_DONATION	Low	3	0.56	0 / 9
1	EDF_DONATION	Medium	3	2.89	0 / 9
1	EDF_DONATION	High	3	7.78	0 / 9
3	RR	Low	1	0.00	0 / 3
3	RR	Medium	1	2.00	0 / 3
3	RR	High	1	16.00	0 / 3
3	EDF	Low	1	0.67	0 / 3
3	EDF	Medium	1	4.67	0 / 3
3	EDF	High	1	8.00	0 / 3
3	EDF_DONATION	Low	1	0.33	0 / 3
3	EDF_DONATION	Medium	1	3.67	0 / 3
3	EDF_DONATION	High	1	10.33	0 / 3



General benchmark elapsed ticks



General benchmark deadline misses

Focused donation demonstration (separate workload)

This controlled test is a four-stage pipe workflow with an unrelated CPU hog. It is not part of sched_bench averages. It was run once for each policy at each CPU count.

CPUs	Policy	Completed	Elapsed ticks	Deadline	Misses	Donation events
1	EDF	1	150	100	1	3
1	EDF_DONATION	1	0	100	0	3
3	EDF	1	150	100	1	3
3	EDF_DONATION	1	0	100	0	3

The focused-run records also retain process IDs and donor/pipe/receiver identifiers in donation_demo_results_separate.csv; raw QEMU logs remain in their original result directories.

Zero elapsed ticks is limited by the xv6 timer resolution and does not mean zero execution time. These single-run observations are not statistical evidence of general policy superiority.

Deliverables

- benchmark_results_cpus1.csv and benchmark_results_cpus3.csv: validated combined general benchmark rows.
- benchmark_results_combined.csv: both CPU counts with source and repetition fields.
- donation_demo_results_separate.csv: focused test data, separate from general benchmark data.
- general_elapsed_ticks.png, general_deadline_misses.png, and focused_donation_comparison.png.
- Original per-run raw logs and CSVs remain preserved under the source directories listed above.